5. Database Schema — TCC Event Clearance App

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A PostgreSQL schema (Supabase) with 19 tables. Event design lives in per-event template data, so changing next event's card never touches old cards.

People tables

Table	Key columns	Notes
departments	id, code (CIT), name, active	Eight rows: CBA, CIT, CED, CCJPS, CHM, CAS, COM, CLIS
sections	id, department_id, year_level, name (3A), school_year	Each department's SBO manages its own list
profiles	id (same as the auth user), email, role (student, officer, head), created_at	One row per Google account
students	id, profile_id, school_id_no, full_name, department_id, section_id, photo_path, thumb_path, photo_status (accepted, needs_retake), status (pending, approved, rejected), rejection_reason, consent_at, school_year	school_id_no is unique
student_devices	id, student_id, public_key, status (active, revoked), created_at, revoked_at, revoked_by	One active device per student
device_reset_requests	id, student_id, new_public_key, new_photo_path, status, reviewed_by, reviewed_at, reason	Approved by the department SBO

Table	Key columns	Notes
officers	id, profile_id, department_id (empty for head), position (for example SBO Auditor), is_head, signature_path, active	The signature image is drawn on the signature pad
scanner_keys	id, officer_id, public_key, device_label, issued_at, revoked_at	One per scanner phone, revocable

Event tables

Table	Key columns	Notes
events	id, name, description, school_year, status (draft, published, archived), location_check (off, flag, block), approved_by_name, approved_on, created_by, published_by, version	version increases on every change so devices know to sync
event_departments	event_id, department_id	Which departments the event is for; empty means all
event_days	id, event_id, day_no, date	One row per day
venues	id, event_id, name, lat, lng, radius_m, location_check_enabled	A pin set by the SBO; an event can have several
slots	id, event_day_id, order_no, label (Time In, Time Out or custom), start_time, end_time, venue_id (optional), signer_officer_id, signer_name, signer_position	Signer name and position are copied in, so old cards never change
card_templates	id, event_id, version, layout_json, artwork_path, created_by	layout_json holds every element of the card: header, logos, fields, signature boxes, colors, fonts, positions
artwork_generations	id, event_id, prompt, image_path, chosen, created_by, created_at	Counted against the regeneration limit per event

The card is never generalized: each event owns its own card_templates row, and a new event can start from a copy of an old one.

Attendance, disputes and audit

Table	Key columns	Notes
attendance	id, client_uuid, event_id, slot_id, student_id, venue_id, officer_id, scanner_key_id, scanned_at, method (qr, manual), location_flag (none, ok, far, unknown), distance_m, status (valid, duplicate, voided), serial_no, signature, note, synced_at	signature is the scanner's signature over the record; this row is the proof behind every stamp
photo_rejections	id, student_id, attendance_id, rejected_by, reason, created_at	Each rejection sets the student's photo_status to needs_retake
disputes	id, student_id, slot_id, note, status (open, resolved), resolved_by, resolution, created_at, resolved_at	For "I was there but the slot is missing"
audit_logs	id, actor_profile_id, action, target_type, target_id, details (JSON), created_at	Append-only: scans, rejections, manual check-ins, resets, edits, publishes

Constraints, indexes and access rules

Constraints

- students.school_id_no is unique.
- student_devices: one active row per student.
- attendance: unique (slot_id, student_id) where status is valid; unique client_uuid; unique serial_no.
- sections: unique (department_id, year_level, name, school_year).
- attendance rows are never edited; only status changes (to voided or duplicate), and each change writes an audit_logs row.

Indexes: attendance (event_id, slot_id), attendance (student_id), students (department_id, section_id), events (status, school_year).

Row-level access

Data	Student	Department officer	Head officer
students, devices, reset requests	Own rows	Own department	All
attendance	Own rows, read only	Own department's students	All
attendance writes	None	Only through the sync function, which checks the scanner signature	Same
events, slots, venues, templates	Published events for the student's department, read only	Read; edit drafts	All; publish
officers, scanner_keys	None	Own rows	All
audit_logs	None	Own department, read only	All, read only

Retention: at the end of each school year, delete photos and archive or delete scan logs, as stated on the consent screen.